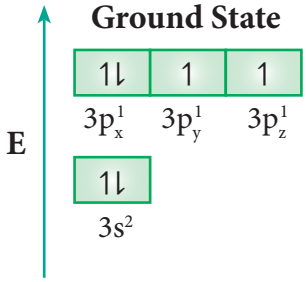
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### $sp^3d^2$ Hybridisation:

In sulphur hexafluoride ($SF_6$) the central atom sulphur extend its octet to undergo $sp^3d^2$ hybridisation to generate six $sp^3d^2$ hybridised orbitals which accounts for six equivalent S-F bonds. The ground state electronic configuration of sulphur is $[Ne]3s^2\ 3p_x^1\ 3p_y^1\ 3p_z^1$.




One electron each from 3s orbital and 3p orbital of sulphur is promoted to its two vacant 3d orbitals ($d_{z^2}$ and $d_{x^2-y^2}$) in the excite state. A total of six valence orbitals from sulphur (one 3s orbital, three 3p orbitals and two 3d orbitals) mixes to give six equivalent $sp^3d^2$ hybridised orbitals. The orbital geometry is octahedral as shown in the figure.

### Overlap with 2pz orbitals of fluorine:

The six $sp^3d^2$ hybridised orbitals of sulphur overlaps linearly with $2p_z$ orbitals of six fluorine atoms to form the six S-F bonds in the sulphur hexafluoride molecule.

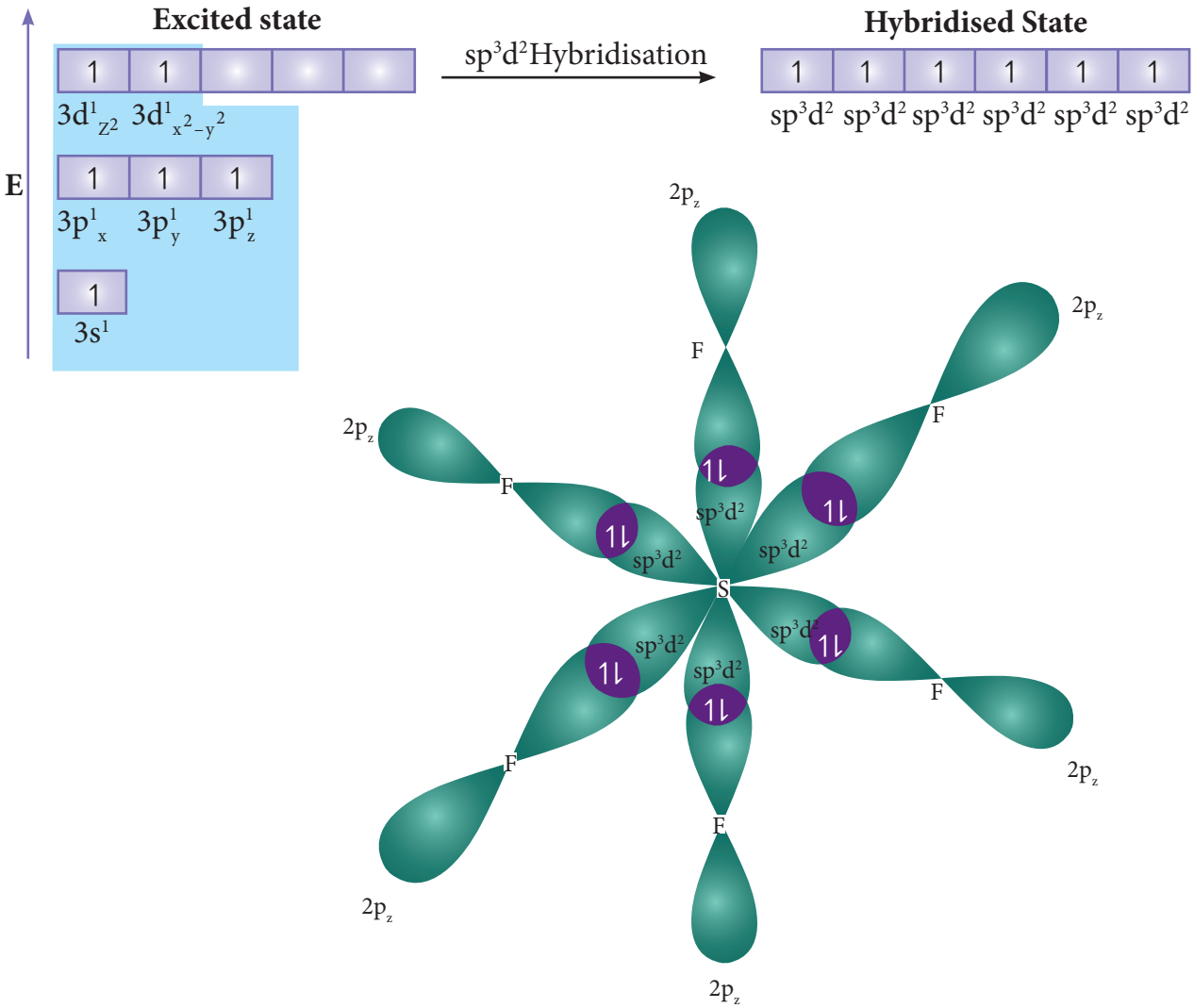



Fig 10.26 $sp^3d^2$ Hybridisation : $SF_6$